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# Data Wrangling II Practical  
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# Cell 1 : Import Required Libraries  
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```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
print("Libraries Imported Successfully")
```

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# -----  
# Cell 2 : Load Dataset  
# -----
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```
df = pd.read_csv('StudentsPerformance.csv')
```

```
print("Dataset Loaded Successfully")  
print(df.head())
```

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# -----  
# Cell 3 : Dataset Information  
# -----
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```
print("\nDataset Information")  
print(df.info())
```

```
print("\nDataset Shape :", df.shape)
```

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# -----  
# Cell 4 : Check Missing Values  
# -----
```

```
print("\nMissing Values")  
print(df.isnull().sum())
```

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# -----
# Cell 5 : Check Duplicate Records
# -----

print("\nDuplicate Rows :", df.duplicated().sum())

# -----
# Cell 6 : Handle Missing Values
# -----

# Numeric columns
num_cols = df.select_dtypes(include=np.number).columns

# Fill numeric missing values with mean
for col in num_cols:
    df[col] = df[col].fillna(df[col].mean())

# Categorical columns
cat_cols = df.select_dtypes(include='object').columns

# Fill categorical missing values with mode
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Handling")
print(df.isnull().sum())

# -----
# Cell 7 : Remove Duplicate Rows
# -----

df.drop_duplicates(inplace=True)

print("\nDataset Shape After Removing Duplicates :", df.shape)

# -----
# Cell 8 : Statistical Summary
# -----

print("\nStatistical Summary")
print(df.describe())

```

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# -----  
# Cell 9 : Distribution Before Outlier Removal  
# -----  
  
df[num_cols].hist(figsize=(10,5), bins=15)  
  
plt.suptitle("Distribution Before Outlier Removal")  
  
plt.show()
```

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# -----  
# Cell 10 : Boxplot for Outlier Detection  
# -----  
  
plt.figure(figsize=(8,5))  
  
sns.boxplot(data=df[num_cols])  
  
plt.title("Boxplot Before Outlier Removal")  
  
plt.xticks(rotation=30)  
  
plt.show()
```

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# -----  
# Cell 11 : Detect Outliers Using IQR  
# -----  
  
print("\nOutlier Detection Using IQR Method")  
  
for col in num_cols:  
  
    Q1 = df[col].quantile(0.25)  
    Q3 = df[col].quantile(0.75)  
  
    IQR = Q3 - Q1  
  
    lower = Q1 - 1.5 * IQR  
    upper = Q3 + 1.5 * IQR
```

```

outliers = df[(df[col] < lower) | (df[col] > upper)]

print(f"\nColumn : {col}")
print(f"Lower Bound : {lower}")
print(f"Upper Bound : {upper}")
print(f"Number of Outliers : {len(outliers)}")

# -----
# Cell 12 : Remove Outliers
# -----

for col in num_cols:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df = df[(df[col] >= lower) & (df[col] <= upper)]

print("\nShape After Outlier Removal :", df.shape)

# -----
# Cell 13 : Distribution After Outlier Removal
# -----

df[num_cols].hist(figsize=(10,5), bins=15)

plt.suptitle("Distribution After Outlier Removal")

plt.show()

# -----
# Cell 14 : Boxplot After Outlier Removal
# -----

plt.figure(figsize=(8,5))

```

```
sns.boxplot(data=df[num_cols])

plt.title("Boxplot After Outlier Removal")

plt.xticks(rotation=30)

plt.show()
```

```
# -----
# Cell 15 : Data Transformation
# -----

# Applying Log Transformation
# Purpose:
# Reduce skewness and make distribution closer to normal distribution
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```
df['math_log'] = np.log1p(df['math score'])

print("\nLog Transformation Applied on 'math score'")
```

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# -----
# Cell 16 : Distribution After Transformation
# -----
```

```
plt.figure(figsize=(6,4))

plt.hist(df['math_log'], bins=15)

plt.title("Distribution After Log Transformation")

plt.xlabel("math_log")

plt.ylabel("Frequency")

plt.show()
```

```
# -----
# Cell 17 : Before vs After Transformation
# -----
```

```
plt.figure(figsize=(10,4))
```

```
# Original Data
plt.subplot(1,2,1)

plt.hist(df['math score'], bins=15)

plt.title("Original Math Score")

# Transformed Data
plt.subplot(1,2,2)

plt.hist(df['math_log'], bins=15)

plt.title("Log Transformed Math Score")

plt.show()


# -----
# Cell 18 : Final Dataset
# -----

print("\nFinal Cleaned Dataset")

print(df.head())


# -----
# Cell 19 : Final Dataset Shape
# -----

print("\nFinal Dataset Shape :", df.shape)
```